Course Title: Cryptography and Network Security Lab	**Course Code**: BCSE309P
Faculty: Prof. Karthika V	Slot: *L15+L16*
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EXERCISE 8- SHA 512

Code

```
import java.util.Scanner;

public class sha512 {

    // ---------- Constants (first 80 primes cube roots) ----------
    private static final long[] K = {
        0x428a2f98d728ae22L,0x7137449123ef65cdL,0xb5c0fbcfec4d3b2fL,0xe9b5dba58189db
bcL,
        0x3956c25bf348b538L,0x59f111f1b605d019L,0x923f82a4af194f9bL,0xab1c5ed5da6d81
18L,
        0xd807aa98a3030242L,0x12835b0145706fbeL,0x243185be4ee4b28cL,0x550c7dc3d5ffb4
e2L,
        0x72be5d74f27b896fL,0x80deb1fe3b1696b1L,0x9bdc06a725c71235L,0xc19bf174cf6926
94L,
        0xe49b69c19ef14ad2L,0xefbe4786384f25e3L,0x0fc19dc68b8cd5b5L,0x240ca1cc77ac9c
65L,
        0x2de92c6f592b0275L,0x4a7484aa6ea6e483L,0x5cb0a9dcbd41fbd4L,0x76f988da831153
b5L,
        0x983e5152ee66dfabL,0xa831c66d2db43210L,0xb00327c898fb213fL,0xbf597fc7beef0e
e4L,
        0xc6e00bf33da88fc2L,0xd5a79147930aa725L,0x06ca6351e003826fL,0x142929670a0e6e
70L,
        0x27b70a8546d22ffcL,0x2e1b21385c26c926L,0x4d2c6dfc5ac42aedL,0x53380d139d95b3
dfL,
        0x650a73548baf63deL,0x766a0abb3c77b2a8L,0x81c2c92e47edaee6L,0x92722c85148235
3bL,
        0xa2bfe8a14cf10364L,0xa81a664bbc423001L,0xc24b8b70d0f89791L,0xc76c51a30654be
30L,
        0xd192e819d6ef5218L,0xd69906245565a910L,0xf40e35855771202aL,0x106aa07032bbd1
b8L,
        0x19a4c116b8d2d0c8L,0x1e376c085141ab53L,0x2748774cdf8eeb99L,0x34b0bcb5e19b48
a8L,
        0x391c0cb3c5c95a63L,0x4ed8aa4ae3418acbL,0x5b9cca4f7763e373L,0x682e6ff3d6b2b8
a3L,
        0x748f82ee5defb2fcL,0x78a5636f43172f60L,0x84c87814a1f0ab72L,0x8cc702081a6439
ecL,
```

```
        0x90befffa23631e28L,0xa4506cebde82bde9L,0xbef9a3f7b2c67915L,0xc67178f2e37253
2bL,
        0xca273eceea26619cL,0xd186b8c721c0c207L,0xeada7dd6cde0eb1eL,0xf57d4f7fee6ed1
78L,
        0x06f067aa72176fbaL,0x0a637dc5a2c898a6L,0x113f9804bef90daeL,0x1b710b35131c47
1bL,
        0x28db77f523047d84L,0x32caab7b40c72493L,0x3c9ebe0a15c9bebcL,0x431d67c49c100d
4cL,
        0x4cc5d4becb3e42b6L,0x597f299cfc657e2aL,0x5fcb6fab3ad6faecL,0x6c44198c4a4758
17L
    };

    // ---------- Rotate functions ----------
    static long ROTR(long x, int n) {
        return (x >>> n) | (x << (64 - n));
    }

    static long Ch(long x, long y, long z) {
        return (x & y) ^ (~x & z);
    }

    static long Maj(long x, long y, long z) {
        return (x & y) ^ (x & z) ^ (y & z);
    }

    static long Sigma0(long x) {
        return ROTR(x,28) ^ ROTR(x,34) ^ ROTR(x,39);
    }

    static long Sigma1(long x) {
        return ROTR(x,14) ^ ROTR(x,18) ^ ROTR(x,41);
    }

    static long sigma0(long x) {
        return ROTR(x,1) ^ ROTR(x,8) ^ (x >>> 7);
    }

    static long sigma1(long x) {
        return ROTR(x,19) ^ ROTR(x,61) ^ (x >>> 6);
    }

    public static void main(String[] args) throws Exception {

        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Input: ");
        String input = sc.nextLine();

        System.out.println("No of characters in input: " + input.length());
```

```java
        byte[] msg = input.getBytes("UTF-8");
        long bitLen = (long) msg.length * 8;

        // ---------- Padding ----------
        int padLen = (112 - (msg.length + 1) % 128 + 128) % 128;
        byte[] padded = new byte[msg.length + 1 + padLen + 16];

        System.arraycopy(msg, 0, padded, 0, msg.length);
        padded[msg.length] = (byte) 0x80;

        for (int i = 0; i < 8; i++) {
            padded[padded.length - 1 - i] = (byte)(bitLen >>> (8 * i));
        }

        // ---------- Initial Hash ----------
        long[] H = {
            0x6a09e667f3bcc908L,0xbb67ae8584caa73bL,
            0x3c6ef372fe94f82bL,0xa54ff53a5f1d36f1L,
            0x510e527fade682d1L,0x9b05688c2b3e6c1fL,
            0x1f83d9abfb41bd6bL,0x5be0cd19137e2179L
        };

        int blocks = padded.length / 128;

        for (int b = 0; b < blocks; b++) {

            System.out.println("\n============================================
Processing Block " + (b + 1) + " ============================================");

            long[] W = new long[80];

            for (int i = 0; i < 16; i++) {
                int idx = b * 128 + i * 8;
                W[i] = 0;
                for (int j = 0; j < 8; j++) {
                    W[i] = (W[i] << 8) | (padded[idx + j] & 0xff);
                }
            }

            for (int i = 16; i < 80; i++) {
                W[i] = sigma1(W[i-2]) + W[i-7] + sigma0(W[i-15]) + W[i-16];
            }

            long a=H[0], b1=H[1], c=H[2], d=H[3];
            long e=H[4], f=H[5], g=H[6], h=H[7];

            // Print Header for the rounds
```

```
            System.out.println("Round
|        A                 B                 C                 D                 E
              F                 G                 H");
            System.out.println("-----------------------------------------------------
-----------------------------------------------------------------------------------
--------------");

            for (int i = 0; i < 80; i++) {

                long T1 = h + Sigma1(e) + Ch(e,f,g) + K[i] + W[i];
                long T2 = Sigma0(a) + Maj(a,b1,c);

                h = g;
                g = f;
                f = e;
                e = d + T1;
                d = c;
                c = b1;
                b1 = a;
                a = T1 + T2;

                // Print the variables after they are updated for this round
                System.out.printf("  %02d  |
%016x  %016x  %016x  %016x  %016x  %016x  %016x  %016x\n",
                                  i, a, b1, c, d, e, f, g, h);
            }

            H[0]+=a; H[1]+=b1; H[2]+=c; H[3]+=d;
            H[4]+=e; H[5]+=f; H[6]+=g; H[7]+=h;
        }

        System.out.print("\nFinal Hash Value: ");
        for (long val : H) {
            System.out.printf("%016x", val);
        }
        System.out.println();
    }
}
```

Output [screenshot]

Input Test Case 1: <896 bits(<112 characters)

```
PS E:\aa vit sem6\z_Crypto-lab\Ex8_sha512> cd "e:\aa vit sem6\z_Crypto-lab\Ex8_sha512\" ; if ($?) { javac sha512.java } ; if ($?) { java sha512 }
Enter Input: arkitabarua
No of characters in input: 11

========================================== Processing Block 1 ==========================================
Round |        A                B                C                D                E                F                G                H
-----------------------------------------------------------------------------------------------------------------------------------------------
  00  | f6bfd6a2315e4056  6a09e667f3bcc908  bb67ae8584caa73b  3c6ef372fe94f82b  58db0a1def1681f2  510e527fade682d1  9b05688c2b3e6c1f  1f83d9abfb41bd6b
  01  | 578271170ddb5b6f  f6bfd6a2315e4056  6a09e667f3bcc908  bb67ae8584caa73b  bbf053e7ded1b25a  58db0a1def1681f2  510e527fade682d1  9b05688c2b3e6c1f
  02  | a71c5d1e22447021  578271170ddb5b6f  f6bfd6a2315e4056  6a09e667f3bcc908  bb1f3ee401c84437  bbf053e7ded1b25a  58db0a1def1681f2  510e527fade682d1
  03  | 329913c4cae6b7b1  a71c5d1e22447021  578271170ddb5b6f  f6bfd6a2315e4056  14720e2441f83233  bb1f3ee401c84437  bbf053e7ded1b25a  58db0a1def1681f2
  04  | 710401d536d3a3f9  329913c4cae6b7b1  a71c5d1e22447021  578271170ddb5b6f  1ae46bd156192694  14720e2441f83233  bb1f3ee401c84437  bbf053e7ded1b25a
  05  | c0ba67bcb8162d53  710401d536d3a3f9  329913c4cae6b7b1  a71c5d1e22447021  5a3d56d0e447bfec  1ae46bd156192694  14720e2441f83233  bb1f3ee401c84437
  06  | 2aace02a0655afe9  c0ba67bcb8162d53  710401d536d3a3f9  329913c4cae6b7b1  8b19c74111b9e12e  5a3d56d0e447bfec  1ae46bd156192694  14720e2441f83233
  07  | 4874ce0fc680fcd4  2aace02a0655afe9  c0ba67bcb8162d53  710401d536d3a3f9  699e41e128dfa122  8b19c74111b9e12e  5a3d56d0e447bfec  1ae46bd156192694
  08  | 795a9b408853f7e1  4874ce0fc680fcd4  2aace02a0655afe9  c0ba67bcb8162d53  1b7ec35efb51b326  699e41e128dfa122  8b19c74111b9e12e  5a3d56d0e447bfec
  09  | 93c4291b10b3a596  795a9b408853f7e1  4874ce0fc680fcd4  2aace02a0655afe9  c5c6a2cd396a2b1a  1b7ec35efb51b326  699e41e128dfa122  8b19c74111b9e12e
  10  | 6b67863d7f8188f8  93c4291b10b3a596  795a9b408853f7e1  4874ce0fc680fcd4  438804144faf2f68  c5c6a2cd396a2b1a  1b7ec35efb51b326  699e41e128dfa122
  11  | ed4cc60bd2daedab  6b67863d7f8188f8  93c4291b10b3a596  795a9b408853f7e1  dd731a592450f73b  438804144faf2f68  c5c6a2cd396a2b1a  1b7ec35efb51b326
  12  | 5777c67e07ed4404  ed4cc60bd2daedab  6b67863d7f8188f8  93c4291b10b3a596  16d00793c567c578  dd731a592450f73b  438804144faf2f68  c5c6a2cd396a2b1a
  13  | 3be2d602e8c38463  5777c67e07ed4404  ed4cc60bd2daedab  6b67863d7f8188f8  5d1e6d12379de35e  16d00793c567c578  dd731a592450f73b  438804144faf2f68
  14  | a3ca3614744cc7e1  3be2d602e8c38463  5777c67e07ed4404  ed4cc60bd2daedab  5bf1d6237a607cb4  5d1e6d12379de35e  16d00793c567c578  dd731a592450f73b
  15  | 9d7b46ea1dcf0c84  a3ca3614744cc7e1  3be2d602e8c38463  5777c67e07ed4404  e5ad6071f4c5bd20  5bf1d6237a607cb4  5d1e6d12379de35e  16d00793c567c578
  16  | df1211b98c5c3837  9d7b46ea1dcf0c84  a3ca3614744cc7e1  3be2d602e8c38463  eae6ecc31da9adc4  e5ad6071f4c5bd20  5bf1d6237a607cb4  5d1e6d12379de35e
  17  | e4251a2756a09466  df1211b98c5c3837  9d7b46ea1dcf0c84  a3ca3614744cc7e1  aae1a471b9bc504f  eae6ecc31da9adc4  e5ad6071f4c5bd20  5bf1d6237a607cb4
  18  | 8ea5cba3f5c7ab10  e4251a2756a09466  df1211b98c5c3837  9d7b46ea1dcf0c84  6430ccae16b7a803  aae1a471b9bc504f  eae6ecc31da9adc4  e5ad6071f4c5bd20
  19  | 1c3dcccf502d77d7  8ea5cba3f5c7ab10  e4251a2756a09466  df1211b98c5c3837  04b3237ab04f48e5  6430ccae16b7a803  aae1a471b9bc504f  eae6ecc31da9adc4
  20  | b156ecd1fa5f1cd7  1c3dcccf502d77d7  8ea5cba3f5c7ab10  e4251a2756a09466  bbc6b9788729d959  04b3237ab04f48e5  6430ccae16b7a803  aae1a471b9bc504f
  21  | aa636deb04d848e9  b156ecd1fa5f1cd7  1c3dcccf502d77d7  8ea5cba3f5c7ab10  79c006d714cfc186  bbc6b9788729d959  04b3237ab04f48e5  6430ccae16b7a803
  22  | e450834ef485abc6  aa636deb04d848e9  b156ecd1fa5f1cd7  1c3dcccf502d77d7  5fc335f679f536ee  79c006d714cfc186  bbc6b9788729d959  04b3237ab04f48e5
  23  | bb251ec4c6002671  e450834ef485abc6  aa636deb04d848e9  b156ecd1fa5f1cd7  ce7d1fffaf7be047  5fc335f679f536ee  79c006d714cfc186  bbc6b9788729d959
```

```
  23  | bb251ec4c6002671  e450834ef485abc6  aa636deb04d848e9  b156ecd1fa5f1cd7  ce7d1fffaf7be047  5fc335f679f536ee  79c006d714cfc186  bbc6b9788729d959
  24  | 87059a2c8bf7233b  bb251ec4c6002671  e450834ef485abc6  aa636deb04d848e9  b5ed086481e73371  ce7d1fffaf7be047  5fc335f679f536ee  79c006d714cfc186
  25  | 792d88ac8cb869f5  87059a2c8bf7233b  bb251ec4c6002671  e450834ef485abc6  b7f34719e474bff4  b5ed086481e73371  ce7d1fffaf7be047  5fc335f679f536ee
  26  | a46b8a2f7de40ac9  792d88ac8cb869f5  87059a2c8bf7233b  bb251ec4c6002671  1be47cd8bd57df2c  b7f34719e474bff4  b5ed086481e73371  ce7d1fffaf7be047
  27  | 36fff8e1ac51fb64  a46b8a2f7de40ac9  792d88ac8cb869f5  87059a2c8bf7233b  ed35273c27735e74  1be47cd8bd57df2c  b7f34719e474bff4  b5ed086481e73371
  28  | 78dd6c361c5f6cab  36fff8e1ac51fb64  a46b8a2f7de40ac9  792d88ac8cb869f5  5e2012485138942e  ed35273c27735e74  1be47cd8bd57df2c  b7f34719e474bff4
  29  | 2aac4be271edd7a1  78dd6c361c5f6cab  36fff8e1ac51fb64  a46b8a2f7de40ac9  40003d23bd40b3f9  5e2012485138942e  ed35273c27735e74  1be47cd8bd57df2c
  30  | 8f0254ea3b918ec0  2aac4be271edd7a1  78dd6c361c5f6cab  36fff8e1ac51fb64  b22aa1e1dadca5a0  40003d23bd40b3f9  5e2012485138942e  ed35273c27735e74
  31  | 695c5ad22bd8b4fe  8f0254ea3b918ec0  2aac4be271edd7a1  78dd6c361c5f6cab  92445a2c4b510291  b22aa1e1dadca5a0  40003d23bd40b3f9  5e2012485138942e
  32  | a37e3274fd496e6e  695c5ad22bd8b4fe  8f0254ea3b918ec0  2aac4be271edd7a1  5e247017ea8f4116  92445a2c4b510291  b22aa1e1dadca5a0  40003d23bd40b3f9
  33  | 04ade059d4a0b52b  a37e3274fd496e6e  695c5ad22bd8b4fe  8f0254ea3b918ec0  81bda99cc83a8628  5e247017ea8f4116  92445a2c4b510291  b22aa1e1dadca5a0
  34  | 2c14544120e47e78  04ade059d4a0b52b  a37e3274fd496e6e  695c5ad22bd8b4fe  0d0ff84980b1327f  81bda99cc83a8628  5e247017ea8f4116  92445a2c4b510291
  35  | e0c1b1350c69f416  2c14544120e47e78  04ade059d4a0b52b  a37e3274fd496e6e  61a26ad50949b100  0d0ff84980b1327f  81bda99cc83a8628  5e247017ea8f4116
  36  | 569fff79ac4cd383  e0c1b1350c69f416  2c14544120e47e78  04ade059d4a0b52b  e5fa521a0aca5138  61a26ad50949b100  0d0ff84980b1327f  81bda99cc83a8628
  37  | 1160bc0310ede0db  569fff79ac4cd383  e0c1b1350c69f416  2c14544120e47e78  54f21178fe2c683a  e5fa521a0aca5138  61a26ad50949b100  0d0ff84980b1327f
  38  | e832f57f53361c8b  1160bc0310ede0db  569fff79ac4cd383  e0c1b1350c69f416  f6c0de49033b7c27  54f21178fe2c683a  e5fa521a0aca5138  61a26ad50949b100
  39  | fdf2fb372e47a2b1  e832f57f53361c8b  1160bc0310ede0db  569fff79ac4cd383  75778b4bbb5136fc  f6c0de49033b7c27  54f21178fe2c683a  e5fa521a0aca5138
  40  | c6cf90c2b950dc63  fdf2fb372e47a2b1  e832f57f53361c8b  1160bc0310ede0db  e245460e8f8d2702  75778b4bbb5136fc  f6c0de49033b7c27  54f21178fe2c683a
  41  | 508f30c43d4d2c67  c6cf90c2b950dc63  fdf2fb372e47a2b1  e832f57f53361c8b  b6d1aab3b81cf965  e245460e8f8d2702  75778b4bbb5136fc  f6c0de49033b7c27
  42  | ac8724453c124086  508f30c43d4d2c67  c6cf90c2b950dc63  fdf2fb372e47a2b1  6bef51ca3f90d29b  b6d1aab3b81cf965  e245460e8f8d2702  75778b4bbb5136fc
  43  | a368afa7e3d18857  ac8724453c124086  508f30c43d4d2c67  c6cf90c2b950dc63  d873bd5066be5b97  6bef51ca3f90d29b  b6d1aab3b81cf965  e245460e8f8d2702
  44  | 570a4649e1ada3c7  a368afa7e3d18857  ac8724453c124086  508f30c43d4d2c67  f31f6dc7ec97771b  d873bd5066be5b97  6bef51ca3f90d29b  b6d1aab3b81cf965
  45  | 5aa9196a9191474c  570a4649e1ada3c7  a368afa7e3d18857  ac8724453c124086  12bc1425c184116c  f31f6dc7ec97771b  d873bd5066be5b97  6bef51ca3f90d29b
  46  | 906ba38909dcdd99  5aa9196a9191474c  570a4649e1ada3c7  a368afa7e3d18857  8177b0ca3fcf1817  12bc1425c184116c  f31f6dc7ec97771b  d873bd5066be5b97
  47  | 5b49e5266a8cc0dc  906ba38909dcdd99  5aa9196a9191474c  570a4649e1ada3c7  dede3a407b407a31  8177b0ca3fcf1817  12bc1425c184116c  f31f6dc7ec97771b
  48  | 994b491d1ace6c72  5b49e5266a8cc0dc  906ba38909dcdd99  5aa9196a9191474c  1731c9c9d7e49178  dede3a407b407a31  8177b0ca3fcf1817  12bc1425c184116c
  49  | 55abf44d8f6dd17d  994b491d1ace6c72  5b49e5266a8cc0dc  906ba38909dcdd99  46a86bbd3e5d76ed  1731c9c9d7e49178  dede3a407b407a31  8177b0ca3fcf1817
  50  | 934a946683af9a6a  55abf44d8f6dd17d  994b491d1ace6c72  5b49e5266a8cc0dc  bc519ab98d40c964  46a86bbd3e5d76ed  1731c9c9d7e49178  dede3a407b407a31
  51  | 839c2446eed25673  934a946683af9a6a  55abf44d8f6dd17d  994b491d1ace6c72  f68516188811c87c  bc519ab98d40c964  46a86bbd3e5d76ed  1731c9c9d7e49178
  52  | 557d33287419d6a1  839c2446eed25673  934a946683af9a6a  55abf44d8f6dd17d  7ff171f6c0d5fb61  f68516188811c87c  bc519ab98d40c964  46a86bbd3e5d76ed
  53  | d8ed7b074b73622f  557d33287419d6a1  839c2446eed25673  934a946683af9a6a  8eca0eddb41ed91e  7ff171f6c0d5fb61  f68516188811c87c  bc519ab98d40c964
  54  | f24ab059c974d0b7  d8ed7b074b73622f  557d33287419d6a1  839c2446eed25673  481bf507b773e0bb  8eca0eddb41ed91e  7ff171f6c0d5fb61  f68516188811c87c
```

```
  54 | f24ab059c974d0b7  d8ed7b074b73622f  557d33287419d6a1  839c2446eed25673  481bf507b773e0bb  8eca0eddb41ed91e  7ff171f6c0d5fb61  f68516188811c87c
  55 | 9a1ae07e57f6a4e7  f24ab059c974d0b7  d8ed7b074b73622f  557d33287419d6a1  f6c6fac84579ebc9  481bf507b773e0bb  8eca0eddb41ed91e  7ff171f6c0d5fb61
  56 | b3ee619a7ad2ec56  9a1ae07e57f6a4e7  f24ab059c974d0b7  d8ed7b074b73622f  18e89a5a1b595816  f6c6fac84579ebc9  481bf507b773e0bb  8eca0eddb41ed91e
  57 | 8f48e7bca063925a  b3ee619a7ad2ec56  9a1ae07e57f6a4e7  f24ab059c974d0b7  ae7ba6c35065b290  18e89a5a1b595816  f6c6fac84579ebc9  481bf507b773e0bb
  58 | 8b2815f29465c5f1  8f48e7bca063925a  b3ee619a7ad2ec56  9a1ae07e57f6a4e7  8ac6de7308a51e68  ae7ba6c35065b290  18e89a5a1b595816  f6c6fac84579ebc9
  59 | 511a8cd0cdbbdd81  8b2815f29465c5f1  8f48e7bca063925a  b3ee619a7ad2ec56  595f21a541f1b398  8ac6de7308a51e68  ae7ba6c35065b290  18e89a5a1b595816
  60 | dd61bddcf7984451  511a8cd0cdbbdd81  8b2815f29465c5f1  8f48e7bca063925a  bcff40b8a6baffb5  595f21a541f1b398  8ac6de7308a51e68  ae7ba6c35065b290
  61 | 5d7b6ade9d3ef80b  dd61bddcf7984451  511a8cd0cdbbdd81  8b2815f29465c5f1  160e504964ef52d1  bcff40b8a6baffb5  595f21a541f1b398  8ac6de7308a51e68
  62 | c8a87ee40d262bb3  5d7b6ade9d3ef80b  dd61bddcf7984451  511a8cd0cdbbdd81  2cbaa4b2ad7e9418  160e504964ef52d1  bcff40b8a6baffb5  595f21a541f1b398
  72 | 093c060b46b54de8  847ae3bd07fdc378  42eba9b6694844e4  11950d32e57935de  e7046fda5cdd81ea  250e95f80002467c  2e053ee1169df4ff  94704c438986aef3
  73 | e2e1c093b3fea397  093c060b46b54de8  847ae3bd07fdc378  42eba9b6694844e4  478742a60fdd7a53  e7046fda5cdd81ea  250e95f80002467c  2e053ee1169df4ff
  74 | 80e40f84bf7f2aee  e2e1c093b3fea397  093c060b46b54de8  847ae3bd07fdc378  4ee489c32767f13c  478742a60fdd7a53  e7046fda5cdd81ea  250e95f80002467c
  75 | 8d10ccb60ab94732  80e40f84bf7f2aee  e2e1c093b3fea397  093c060b46b54de8  bf540fe8e73fab07  4ee489c32767f13c  478742a60fdd7a53  e7046fda5cdd81ea
  76 | 0d01500c69c1b828  8d10ccb60ab94732  80e40f84bf7f2aee  e2e1c093b3fea397  502d3916dd250886  bf540fe8e73fab07  4ee489c32767f13c  478742a60fdd7a53
  77 | 4d6099f7b157e3b3  0d01500c69c1b828  8d10ccb60ab94732  80e40f84bf7f2aee  04899e0cb60e06bb  502d3916dd250886  bf540fe8e73fab07  4ee489c32767f13c
  78 | 6c5dd4a82ad42b01  4d6099f7b157e3b3  0d01500c69c1b828  8d10ccb60ab94732  c9f79f575cb5fa88  04899e0cb60e06bb  502d3916dd250886  bf540fe8e73fab07
  79 | 26d920aa00056b14  6c5dd4a82ad42b01  4d6099f7b157e3b3  0d01500c69c1b828  6f060a3344d68324  c9f79f575cb5fa88  04899e0cb60e06bb  502d3916dd250886

Final Hash Value: 90e30711f3c2341c27c5832daf9ed23c89cf8d6aafecdbdeb2514546c8deef19c0145cb2f2bd05f564fd07e387f466a7240d77b8b14fc426ac0e062ff0a329ff
```

Input Test Case 2: =896 bits(=112 characters)

```
PS E:\aa vit sem6\z_Crypto-lab\Ex8_sha512> cd "e:\aa vit sem6\z_Crypto-lab\Ex8_sha512\" ; if ($?) { javac sha512.java } ; if ($?) { java sha512 }
Enter Input: arkitabaruaarkitabaruaarkitabaruaarkitabaruaarkitabaruaarkitabaruaarkitabaruacryptographycryptography
No of characters in input: 112

========================================== Processing Block 1 ==========================================
Round |        A                 B                 C                 D                 E                 F                 G
        H
-----------------------------------------------------------------------------------------------------------------------------------------
  00  | f6bfd6a2315e4056  6a09e667f3bcc908  bb67ae8584caa73b  3c6ef372fe94f82b  58db0a1def1681f2  510e527fade682d1  9b05688c2b3e6c1f  1f83d9abfb41bd6b
  01  | 578270f88046c4e3  f6bfd6a2315e4056  6a09e667f3bcc908  bb67ae8584caa73b  bbf053c9513d1bce  58db0a1def1681f2  510e527fade682d1  9b05688c2b3e6c1f
  02  | 247211265415cabe  578270f88046c4e3  f6bfd6a2315e4056  6a09e667f3bcc908  93d05644bc191a60  bbf053c9513d1bce  58db0a1def1681f2  510e527fade682d1
  03  | 9ef1a7932487d721  247211265415cabe  578270f88046c4e3  f6bfd6a2315e4056  f9574a8055786d25  93d05644bc191a60  bbf053c9513d1bce  58db0a1def1681f2
  04  | 357f739e99f258cc  9ef1a7932487d721  247211265415cabe  578270f88046c4e3  6e37900efbacdee6  f9574a8055786d25  93d05644bc191a60  bbf053c9513d1bce
  05  | f56748716c57e599  357f739e99f258cc  9ef1a7932487d721  247211265415cabe  140b87921544b3e7  6e37900efbacdee6  f9574a8055786d25  93d05644bc191a60
  06  | e462ad8780aaf0e5  f56748716c57e599  357f739e99f258cc  9ef1a7932487d721  d6aa4be88f214a80  140b87921544b3e7  6e37900efbacdee6  f9574a8055786d25
  07  | bacff71681f5d80c  e462ad8780aaf0e5  f56748716c57e599  357f739e99f258cc  7ed54d8b2890081e  d6aa4be88f214a80  140b87921544b3e7  6e37900efbacdee6
  08  | 80a09f26ae0017fc  bacff71681f5d80c  e462ad8780aaf0e5  f56748716c57e599  2f9508352bc16cf8  7ed54d8b2890081e  d6aa4be88f214a80  140b87921544b3e7
  09  | 21bb2ef8db463522  80a09f26ae0017fc  bacff71681f5d80c  e462ad8780aaf0e5  6f623e4af3dcb532  2f9508352bc16cf8  7ed54d8b2890081e  d6aa4be88f214a80
  10  | 949256ccbdaac8dd  21bb2ef8db463522  80a09f26ae0017fc  bacff71681f5d80c  6544f21cdb725228  6f623e4af3dcb532  2f9508352bc16cf8  7ed54d8b2890081e
  11  | 66db3dc19dc24e9f  949256ccbdaac8dd  21bb2ef8db463522  80a09f26ae0017fc  343bab6bc98d32da  6544f21cdb725228  6f623e4af3dcb532  2f9508352bc16cf8
  12  | 024a5d3acffac232  66db3dc19dc24e9f  949256ccbdaac8dd  21bb2ef8db463522  25dfbecc556fe87d  343bab6bc98d32da  6544f21cdb725228  6f623e4af3dcb532
  13  | 47357002e3b030e6  024a5d3acffac232  66db3dc19dc24e9f  949256ccbdaac8dd  2848db0a60d7cb11  25dfbecc556fe87d  343bab6bc98d32da  6544f21cdb725228
  14  | ddd113419b0f15c3  47357002e3b030e6  024a5d3acffac232  66db3dc19dc24e9f  a5df8acf5ab3651c  2848db0a60d7cb11  25dfbecc556fe87d  343bab6bc98d32da
  15  | dd0f1b0fa04b5a54  ddd113419b0f15c3  47357002e3b030e6  024a5d3acffac232  a7948067c474c522  a5df8acf5ab3651c  2848db0a60d7cb11  25dfbecc556fe87d
  16  | b2f52714e7630394  dd0f1b0fa04b5a54  ddd113419b0f15c3  47357002e3b030e6  e4428bdfc64a4cf1  a7948067c474c522  a5df8acf5ab3651c  2848db0a60d7cb11
  17  | 6eb51ba2b975f8a6  b2f52714e7630394  dd0f1b0fa04b5a54  ddd113419b0f15c3  71ef38f6cf5042f3  e4428bdfc64a4cf1  a7948067c474c522  a5df8acf5ab3651c
  18  | 24024234226096d6  6eb51ba2b975f8a6  b2f52714e7630394  dd0f1b0fa04b5a54  86ae1ab0dfeabb11  71ef38f6cf5042f3  e4428bdfc64a4cf1  a7948067c474c522
  19  | 459da358392e166b  24024234226096d6  6eb51ba2b975f8a6  b2f52714e7630394  b5223138d0ab26de  86ae1ab0dfeabb11  71ef38f6cf5042f3  e4428bdfc64a4cf1
  20  | 476f75a0c48f57da  459da358392e166b  24024234226096d6  6eb51ba2b975f8a6  68f6da8314575e75  b5223138d0ab26de  86ae1ab0dfeabb11  71ef38f6cf5042f3
  21  | 5358fa4b10907e88  476f75a0c48f57da  459da358392e166b  24024234226096d6  449efbc0563606c5  68f6da8314575e75  b5223138d0ab26de  86ae1ab0dfeabb11
```

22	cc8358a214fef85f	5358fa4b10907e88	476f75a0c48f57da	459da358392e166b	4e24cfb3f4d20c94	449efbc0563606c5	68f6da8314575e75	b5223138d0ab26de
23	c5a3256e432acb78	cc8358a214fef85f	5358fa4b10907e88	476f75a0c48f57da	34fb8a08e32e0851	4e24cfb3f4d20c94	449efbc0563606c5	68f6da8314575e75
24	2121d36350bd210f	c5a3256e432acb78	cc8358a214fef85f	5358fa4b10907e88	242d7fd569bfa49c	34fb8a08e32e0851	4e24cfb3f4d20c94	449efbc0563606c5
25	01c83a3932690a44	2121d36350bd210f	c5a3256e432acb78	cc8358a214fef85f	7621c02e2e239e22	242d7fd569bfa49c	34fb8a08e32e0851	4e24cfb3f4d20c94
26	098ca8dca517d67f	01c83a3932690a44	2121d36350bd210f	c5a3256e432acb78	bc00994dd2fb8629	7621c02e2e239e22	242d7fd569bfa49c	34fb8a08e32e0851
27	5032f2910a9c9eac	098ca8dca517d67f	01c83a3932690a44	2121d36350bd210f	52da9fc278cfa929	bc00994dd2fb8629	7621c02e2e239e22	242d7fd569bfa49c
28	18b08c245c3d341f	5032f2910a9c9eac	098ca8dca517d67f	01c83a3932690a44	6ecdb09b3b58c671	52da9fc278cfa929	bc00994dd2fb8629	7621c02e2e239e22
29	654b57d468967131	18b08c245c3d341f	5032f2910a9c9eac	098ca8dca517d67f	b1fe72db1bcc64e2	6ecdb09b3b58c671	52da9fc278cfa929	bc00994dd2fb8629
30	1c6a853ad1fd5c17	654b57d468967131	18b08c245c3d341f	5032f2910a9c9eac	9a30b3cabf4bc03d	b1fe72db1bcc64e2	6ecdb09b3b58c671	52da9fc278cfa929
31	8baaa76411e657c2	1c6a853ad1fd5c17	654b57d468967131	18b08c245c3d341f	e169a7f4943b5a6e	9a30b3cabf4bc03d	b1fe72db1bcc64e2	6ecdb09b3b58c671
32	c384694be49be0ba	8baaa76411e657c2	1c6a853ad1fd5c17	654b57d468967131	fc8b4884518b34f0	e169a7f4943b5a6e	9a30b3cabf4bc03d	b1fe72db1bcc64e2
33	2765d712abee4149	c384694be49be0ba	8baaa76411e657c2	1c6a853ad1fd5c17	d9b4c5384564d7aa	fc8b4884518b34f0	e169a7f4943b5a6e	9a30b3cabf4bc03d
34	801e7ffaa60ca235	2765d712abee4149	c384694be49be0ba	8baaa76411e657c2	e79bc5b02850ed42	d9b4c5384564d7aa	fc8b4884518b34f0	e169a7f4943b5a6e
35	598959b27bec52d2	801e7ffaa60ca235	2765d712abee4149	c384694be49be0ba	262a6f2adc636db0	e79bc5b02850ed42	d9b4c5384564d7aa	fc8b4884518b34f0
36	57f310261c107262	598959b27bec52d2	801e7ffaa60ca235	2765d712abee4149	d5a0388aaa7b30d3	262a6f2adc636db0	e79bc5b02850ed42	d9b4c5384564d7aa
37	1857312fa1d2e28c	57f310261c107262	598959b27bec52d2	801e7ffaa60ca235	e3e69436e152911b	d5a0388aaa7b30d3	262a6f2adc636db0	e79bc5b02850ed42
38	30a0bc4a47b07ad2	1857312fa1d2e28c	57f310261c107262	598959b27bec52d2	acd2f57718963972	e3e69436e152911b	d5a0388aaa7b30d3	262a6f2adc636db0
39	94f9c5686a1b7413	30a0bc4a47b07ad2	1857312fa1d2e28c	57f310261c107262	5f2b1b8abe34a955	acd2f57718963972	e3e69436e152911b	d5a0388aaa7b30d3
40	341fc79266b33720	94f9c5686a1b7413	30a0bc4a47b07ad2	1857312fa1d2e28c	0f3b777891a5629a	5f2b1b8abe34a955	acd2f57718963972	e3e69436e152911b
41	62f179af68992c6c	341fc79266b33720	94f9c5686a1b7413	30a0bc4a47b07ad2	703c0bef9724e199	0f3b777891a5629a	5f2b1b8abe34a955	acd2f57718963972
42	9c33c577cad16d55	62f179af68992c6c	341fc79266b33720	94f9c5686a1b7413	8a74fc91b8788d99	703c0bef9724e199	0f3b777891a5629a	5f2b1b8abe34a955
43	5da52c7bc6c21d98	9c33c577cad16d55	62f179af68992c6c	341fc79266b33720	0e33ff56b743a29c	8a74fc91b8788d99	703c0bef9724e199	0f3b777891a5629a
44	2475d46beac0e315	5da52c7bc6c21d98	9c33c577cad16d55	62f179af68992c6c	91c753a5892225df	0e33ff56b743a29c	8a74fc91b8788d99	703c0bef9724e199
45	c646013342c50864	2475d46beac0e315	5da52c7bc6c21d98	9c33c577cad16d55	8b962e15bc94efaf	91c753a5892225df	0e33ff56b743a29c	8a74fc91b8788d99
46	95aee0814f4fb9f1	c646013342c50864	2475d46beac0e315	5da52c7bc6c21d98	53195337bae3feb8	8b962e15bc94efaf	91c753a5892225df	0e33ff56b743a29c
47	10835068c09b05b7	95aee0814f4fb9f1	c646013342c50864	2475d46beac0e315	440aceaa5fe88be5	53195337bae3feb8	8b962e15bc94efaf	91c753a5892225df
48	95f2697a5c5bd500	10835068c09b05b7	95aee0814f4fb9f1	c646013342c50864	3dca51ad6218d9ea	440aceaa5fe88be5	53195337bae3feb8	8b962e15bc94efaf
49	80292070f2ac9549	95f2697a5c5bd500	10835068c09b05b7	95aee0814f4fb9f1	0ab9ae586da41ed3	3dca51ad6218d9ea	440aceaa5fe88be5	53195337bae3feb8
50	79969953969e7237	80292070f2ac9549	95f2697a5c5bd500	10835068c09b05b7	871af07b23ba79d4	0ab9ae586da41ed3	3dca51ad6218d9ea	440aceaa5fe88be5
51	600b24d4b507caf0	79969953969e7237	80292070f2ac9549	95f2697a5c5bd500	b36ec8ac7504dd47	871af07b23ba79d4	0ab9ae586da41ed3	3dca51ad6218d9ea
52	54ec6a2c04e346d7	600b24d4b507caf0	79969953969e7237	80292070f2ac9549	b67c6125b23fb72f	b36ec8ac7504dd47	871af07b23ba79d4	0ab9ae586da41ed3
53	8b0f140ed0106f62	54ec6a2c04e346d7	600b24d4b507caf0	79969953969e7237	83a491ddf8e16115	b67c6125b23fb72f	b36ec8ac7504dd47	871af07b23ba79d4
54	095fb241cb7c1363	8b0f140ed0106f62	54ec6a2c04e346d7	600b24d4b507caf0	9a445a59f6f29ae2	83a491ddf8e16115	b67c6125b23fb72f	b36ec8ac7504dd47
55	612e34a5a91c610c	095fb241cb7c1363	8b0f140ed0106f62	54ec6a2c04e346d7	71615c9f0bf56cb1	9a445a59f6f29ae2	83a491ddf8e16115	b67c6125b23fb72f
56	b2bc79e1b72a145b	612e34a5a91c610c	095fb241cb7c1363	8b0f140ed0106f62	4dc67fc0e0865cb6	71615c9f0bf56cb1	9a445a59f6f29ae2	83a491ddf8e16115
57	7a45d8dc48626718	b2bc79e1b72a145b	612e34a5a91c610c	095fb241cb7c1363	08112783fd28cc9f	4dc67fc0e0865cb6	71615c9f0bf56cb1	9a445a59f6f29ae2
58	f458275c7dd88f88	7a45d8dc48626718	b2bc79e1b72a145b	612e34a5a91c610c	5edd342915f1cd91	08112783fd28cc9f	4dc67fc0e0865cb6	71615c9f0bf56cb1
59	17c2e868ad28e936	f458275c7dd88f88	7a45d8dc48626718	b2bc79e1b72a145b	0c8f38bf6f5e76cc	5edd342915f1cd91	08112783fd28cc9f	4dc67fc0e0865cb6
60	e05e8bbfb0ab546c	17c2e868ad28e936	f458275c7dd88f88	7a45d8dc48626718	f43b2446647ac06f	0c8f38bf6f5e76cc	5edd342915f1cd91	08112783fd28cc9f
61	81c5ca82fc6cc2ea	e05e8bbfb0ab546c	17c2e868ad28e936	f458275c7dd88f88	6db23264a2e6b8d3	f43b2446647ac06f	0c8f38bf6f5e76cc	5edd342915f1cd91
62	70f7632017a7cb1c	81c5ca82fc6cc2ea	e05e8bbfb0ab546c	17c2e868ad28e936	6758f83a7029491c	6db23264a2e6b8d3	f43b2446647ac06f	0c8f38bf6f5e76cc
63	b26eeba4f596f627	70f7632017a7cb1c	81c5ca82fc6cc2ea	e05e8bbfb0ab546c	a99ffbd4c26618e2	6758f83a7029491c	6db23264a2e6b8d3	f43b2446647ac06f
64	65968f0995307b41	b26eeba4f596f627	70f7632017a7cb1c	81c5ca82fc6cc2ea	672b3d09cb23300e	a99ffbd4c26618e2	6758f83a7029491c	6db23264a2e6b8d3
65	c534e71a8b1a8dba	65968f0995307b41	b26eeba4f596f627	70f7632017a7cb1c	b0a1fc4c6f09d75a	672b3d09cb23300e	a99ffbd4c26618e2	6758f83a7029491c
66	997564132fb81ae1	c534e71a8b1a8dba	65968f0995307b41	b26eeba4f596f627	febd8a511ac3c53a	b0a1fc4c6f09d75a	672b3d09cb23300e	a99ffbd4c26618e2
67	62a773fac1200ed2	997564132fb81ae1	c534e71a8b1a8dba	65968f0995307b41	39b09feff544f61a	febd8a511ac3c53a	b0a1fc4c6f09d75a	672b3d09cb23300e
68	299b24808a133a78	62a773fac1200ed2	997564132fb81ae1	c534e71a8b1a8dba	76319de07deff912	39b09feff544f61a	febd8a511ac3c53a	b0a1fc4c6f09d75a
69	9248ad76a55cd305	299b24808a133a78	62a773fac1200ed2	997564132fb81ae1	ab22e09641bf8e6e	76319de07deff912	39b09feff544f61a	febd8a511ac3c53a
70	943d291afacff588	9248ad76a55cd305	299b24808a133a78	62a773fac1200ed2	f956a9fd5e3b88ac	ab22e09641bf8e6e	76319de07deff912	39b09feff544f61a
71	e5a530e7fcbf0a37	943d291afacff588	9248ad76a55cd305	299b24808a133a78	907a3dcf9b898446	f956a9fd5e3b88ac	ab22e09641bf8e6e	76319de07deff912
72	2f7317d4b216235b	e5a530e7fcbf0a37	943d291afacff588	9248ad76a55cd305	c9baf2f6725882a7	907a3dcf9b898446	f956a9fd5e3b88ac	ab22e09641bf8e6e
73	187ab7a158362a0f	2f7317d4b216235b	e5a530e7fcbf0a37	943d291afacff588	610aa21e78407b68	c9baf2f6725882a7	907a3dcf9b898446	f956a9fd5e3b88ac
74	30e0ddcd0e7680e3	187ab7a158362a0f	2f7317d4b216235b	e5a530e7fcbf0a37	ffcb88dbb18b29ba	610aa21e78407b68	c9baf2f6725882a7	907a3dcf9b898446
75	2ca726fc267a6330	30e0ddcd0e7680e3	187ab7a158362a0f	2f7317d4b216235b	9af07d1504af2104	ffcb88dbb18b29ba	610aa21e78407b68	c9baf2f6725882a7
76	a36f3bda44c574cf	2ca726fc267a6330	30e0ddcd0e7680e3	187ab7a158362a0f	038b3cb947628dd7	9af07d1504af2104	ffcb88dbb18b29ba	610aa21e78407b68
77	546bcc923d70cf08	a36f3bda44c574cf	2ca726fc267a6330	30e0ddcd0e7680e3	e20fa9374dc1eb0f	038b3cb947628dd7	9af07d1504af2104	ffcb88dbb18b29ba
78	f2457c7c3325f769	546bcc923d70cf08	a36f3bda44c574cf	2ca726fc267a6330	828d0895da1d23a6	e20fa9374dc1eb0f	038b3cb947628dd7	9af07d1504af2104
79	bfa0e140a9edee3f	f2457c7c3325f769	546bcc923d70cf08	a36f3bda44c574cf	32e84ab70fe04802	828d0895da1d23a6	e20fa9374dc1eb0f	038b3cb947628dd7

========== Processing Block 2 ==========

Round	A	B	C	D	E	F	G	H
00	fcd2a960cdd9e192	29aac7a89daab747	adad2b01b7f09ea4	90dac0053c05c733	eeec8aa5d586573f	83f69d36bdc6cad3	1d927122055b8fc5	019382e34903a87a
01	7fbac166907c7ee3	fcd2a960cdd9e192	29aac7a89daab747	adad2b01b7f09ea4	337703c2d6a1f58a	eeec8aa5d586573f	83f69d36bdc6cad3	1d927122055b8fc5
02	ad0be8757b084a72	7fbac166907c7ee3	fcd2a960cdd9e192	29aac7a89daab747	6e0648a3b2421da1	337703c2d6a1f58a	eeec8aa5d586573f	83f69d36bdc6cad3
03	b31b4c80eee4fa75	ad0be8757b084a72	7fbac166907c7ee3	fcd2a960cdd9e192	da7a85a21490dc30	6e0648a3b2421da1	337703c2d6a1f58a	eeec8aa5d586573f
04	bd567f96fafedc70	b31b4c80eee4fa75	ad0be8757b084a72	7fbac166907c7ee3	26e21060185815d8	da7a85a21490dc30	6e0648a3b2421da1	337703c2d6a1f58a
05	33ac16d67e7a9d37	bd567f96fafedc70	b31b4c80eee4fa75	ad0be8757b084a72	b9a3ddf8f8c17735	26e21060185815d8	da7a85a21490dc30	6e0648a3b2421da1
06	6891a3a22f39249b	33ac16d67e7a9d37	bd567f96fafedc70	b31b4c80eee4fa75	8db3ac7c12f56199	b9a3ddf8f8c17735	26e21060185815d8	da7a85a21490dc30
07	871a5a72825b7717	6891a3a22f39249b	33ac16d67e7a9d37	bd567f96fafedc70	c49efc838cd663cc	8db3ac7c12f56199	b9a3ddf8f8c17735	26e21060185815d8
08	9660de24c56566fc	871a5a72825b7717	6891a3a22f39249b	33ac16d67e7a9d37	cffa3075d27c0955	c49efc838cd663cc	8db3ac7c12f56199	b9a3ddf8f8c17735
09	63453df4e867844e	9660de24c56566fc	871a5a72825b7717	6891a3a22f39249b	e25a7ded24afa66d	cffa3075d27c0955	c49efc838cd663cc	8db3ac7c12f56199
10	dc7452403940cd3b	63453df4e867844e	9660de24c56566fc	871a5a72825b7717	68142c7077ce7130	e25a7ded24afa66d	cffa3075d27c0955	c49efc838cd663cc
11	52eed3e85bc4a6a4	dc7452403940cd3b	63453df4e867844e	9660de24c56566fc	e9766e8cacd98106	68142c7077ce7130	e25a7ded24afa66d	cffa3075d27c0955
12	9b525d127902b99a	52eed3e85bc4a6a4	dc7452403940cd3b	63453df4e867844e	65421cb552304490	e9766e8cacd98106	68142c7077ce7130	e25a7ded24afa66d
13	2b6b7726e377bc14	9b525d127902b99a	52eed3e85bc4a6a4	dc7452403940cd3b	89a12f1f2bd6afcf	65421cb552304490	e9766e8cacd98106	68142c7077ce7130
14	de69c7ca43b072fa	2b6b7726e377bc14	9b525d127902b99a	52eed3e85bc4a6a4	dd1301a86d6f0a58	89a12f1f2bd6afcf	65421cb552304490	e9766e8cacd98106
15	6eef1ca5aa2bb8bc	de69c7ca43b072fa	2b6b7726e377bc14	9b525d127902b99a	e7064594fe02c7ed	dd1301a86d6f0a58	89a12f1f2bd6afcf	65421cb552304490
16	6d8520fd41888a5b	6eef1ca5aa2bb8bc	de69c7ca43b072fa	2b6b7726e377bc14	1706f3d049bfb7ef	e7064594fe02c7ed	dd1301a86d6f0a58	89a12f1f2bd6afcf
17	44a099814e384249	6d8520fd41888a5b	6eef1ca5aa2bb8bc	de69c7ca43b072fa	4eb57586381ef5b9	1706f3d049bfb7ef	e7064594fe02c7ed	dd1301a86d6f0a58
18	5badd429f2ca5660	44a099814e384249	6d8520fd41888a5b	6eef1ca5aa2bb8bc	1adc3ecc62a93fba	4eb57586381ef5b9	1706f3d049bfb7ef	e7064594fe02c7ed
19	43798e367768a4e6	5badd429f2ca5660	44a099814e384249	6d8520fd41888a5b	60d0b301728aaf1c	1adc3ecc62a93fba	4eb57586381ef5b9	1706f3d049bfb7ef
20	b65dc71d5ea5f344	43798e367768a4e6	5badd429f2ca5660	44a099814e384249	587a95e4be05c349	60d0b301728aaf1c	1adc3ecc62a93fba	4eb57586381ef5b9
21	626fb478dfcdfa52	b65dc71d5ea5f344	43798e367768a4e6	5badd429f2ca5660	cc47843ff5f58a9b	587a95e4be05c349	60d0b301728aaf1c	1adc3ecc62a93fba
22	d4a39f1b0ca357c5	626fb478dfcdfa52	b65dc71d5ea5f344	43798e367768a4e6	9340acc284db45e4	cc47843ff5f58a9b	587a95e4be05c349	60d0b301728aaf1c
23	b5625b90331adc01	d4a39f1b0ca357c5	626fb478dfcdfa52	b65dc71d5ea5f344	c36763a997253c5b	9340acc284db45e4	cc47843ff5f58a9b	587a95e4be05c349
24	fcfcdd64648fb64d	b5625b90331adc01	d4a39f1b0ca357c5	626fb478dfcdfa52	a1e9c2c66b91e000	c36763a997253c5b	9340acc284db45e4	cc47843ff5f58a9b
25	f1e85c8b7d47f225	fcfcdd64648fb64d	b5625b90331adc01	d4a39f1b0ca357c5	c6639bd40d7efd8d	a1e9c2c66b91e000	c36763a997253c5b	9340acc284db45e4
26	7bb93dbda65cf40f	f1e85c8b7d47f225	fcfcdd64648fb64d	b5625b90331adc01	3ca02e2614d446a3	c6639bd40d7efd8d	a1e9c2c66b91e000	c36763a997253c5b
27	f9670e8b96b4a11d	7bb93dbda65cf40f	f1e85c8b7d47f225	fcfcdd64648fb64d	3dbc4851e9f4a043	3ca02e2614d446a3	c6639bd40d7efd8d	a1e9c2c66b91e000
28	ed499d332b2db3ef	f9670e8b96b4a11d	7bb93dbda65cf40f	f1e85c8b7d47f225	56930d31058c9529	3dbc4851e9f4a043	3ca02e2614d446a3	c6639bd40d7efd8d
29	1a3f1e9017ab537e	ed499d332b2db3ef	f9670e8b96b4a11d	7bb93dbda65cf40f	f47853fcfea56d50	56930d31058c9529	3dbc4851e9f4a043	3ca02e2614d446a3
30	22472d1ee2f37d4c	1a3f1e9017ab537e	ed499d332b2db3ef	f9670e8b96b4a11d	452096b098586d44	f47853fcfea56d50	56930d31058c9529	3dbc4851e9f4a043
31	f0b7899b3ccb71aa	22472d1ee2f37d4c	1a3f1e9017ab537e	ed499d332b2db3ef	15808da8fb2d0a66	452096b098586d44	f47853fcfea56d50	56930d31058c9529
32	20ae4c534edc2a5e	f0b7899b3ccb71aa	22472d1ee2f37d4c	1a3f1e9017ab537e	a5c48bc5606a5819	15808da8fb2d0a66	452096b098586d44	f47853fcfea56d50
33	d1f1c3bbe3e2feec	20ae4c534edc2a5e	f0b7899b3ccb71aa	22472d1ee2f37d4c	32a1be0f4e24ce64	a5c48bc5606a5819	15808da8fb2d0a66	452096b098586d44
34	afabd61967a0f4f7	d1f1c3bbe3e2feec	20ae4c534edc2a5e	f0b7899b3ccb71aa	302aa510e9064e7e	32a1be0f4e24ce64	a5c48bc5606a5819	15808da8fb2d0a66
35	676ac2c3b7a7770d	afabd61967a0f4f7	d1f1c3bbe3e2feec	20ae4c534edc2a5e	a54e52948e8925fd	302aa510e9064e7e	32a1be0f4e24ce64	a5c48bc5606a5819
36	7e703184409a7b42	676ac2c3b7a7770d	afabd61967a0f4f7	d1f1c3bbe3e2feec	a640d740721b65a5	a54e52948e8925fd	302aa510e9064e7e	32a1be0f4e24ce64
37	0119303076d1dd6b	7e703184409a7b42	676ac2c3b7a7770d	afabd61967a0f4f7	52a00368f6ae70ca	a640d740721b65a5	a54e52948e8925fd	302aa510e9064e7e
38	12d5ecb437071b98	0119303076d1dd6b	7e703184409a7b42	676ac2c3b7a7770d	4ac58ffd203d13d9	52a00368f6ae70ca	a640d740721b65a5	a54e52948e8925fd
39	67f055a875c07ef8	12d5ecb437071b98	0119303076d1dd6b	7e703184409a7b42	a72b76679d067f04	4ac58ffd203d13d9	52a00368f6ae70ca	a640d740721b65a5
40	daa0b0b914d17ce4	67f055a875c07ef8	12d5ecb437071b98	0119303076d1dd6b	43a2fcc74773e8e8	a72b76679d067f04	4ac58ffd203d13d9	52a00368f6ae70ca
41	c97f676c54c5e204	daa0b0b914d17ce4	67f055a875c07ef8	12d5ecb437071b98	fd9d6f5740bfd999	43a2fcc74773e8e8	a72b76679d067f04	4ac58ffd203d13d9
42	5c711328526072bd	c97f676c54c5e204	daa0b0b914d17ce4	67f055a875c07ef8	218fb72a886abda1	fd9d6f5740bfd999	43a2fcc74773e8e8	a72b76679d067f04
43	09570fc1320807d5	5c711328526072bd	c97f676c54c5e204	daa0b0b914d17ce4	369a3aa168517fc0	218fb72a886abda1	fd9d6f5740bfd999	43a2fcc74773e8e8
44	f6ab75b2f5912e33	09570fc1320807d5	5c711328526072bd	c97f676c54c5e204	996eb2593aeab686	369a3aa168517fc0	218fb72a886abda1	fd9d6f5740bfd999
45	a5d4c56ca5b6a06d	f6ab75b2f5912e33	09570fc1320807d5	5c711328526072bd	91438a48978b8b14	996eb2593aeab686	369a3aa168517fc0	218fb72a886abda1
46	09a95ff187a581d6	a5d4c56ca5b6a06d	f6ab75b2f5912e33	09570fc1320807d5	14f669b735030003	91438a48978b8b14	996eb2593aeab686	369a3aa168517fc0
47	2a457df84b778d92	09a95ff187a581d6	a5d4c56ca5b6a06d	f6ab75b2f5912e33	953501b222dcfa95	14f669b735030003	91438a48978b8b14	996eb2593aeab686
48	7a239c8496f07571	2a457df84b778d92	09a95ff187a581d6	a5d4c56ca5b6a06d	f1d5dee1fa3817cd	953501b222dcfa95	14f669b735030003	91438a48978b8b14
49	86465f4a1f850f87	7a239c8496f07571	2a457df84b778d92	09a95ff187a581d6	be631c251f00c1d1	f1d5dee1fa3817cd	953501b222dcfa95	14f669b735030003
50	9c17368d5f11d7f9	86465f4a1f850f87	7a239c8496f07571	2a457df84b778d92	8fea87303bc95c6f	be631c251f00c1d1	f1d5dee1fa3817cd	953501b222dcfa95
51	276abdeaeae9850f	9c17368d5f11d7f9	86465f4a1f850f87	7a239c8496f07571	f741f38ec1812f95	8fea87303bc95c6f	be631c251f00c1d1	f1d5dee1fa3817cd
52	fecd331edbf976c5	276abdeaeae9850f	9c17368d5f11d7f9	86465f4a1f850f87	30b2ad1d7228bff8	f741f38ec1812f95	8fea87303bc95c6f	be631c251f00c1d1
53	dea048283fd97502	fecd331edbf976c5	276abdeaeae9850f	9c17368d5f11d7f9	71b8ace06ac6c570	30b2ad1d7228bff8	f741f38ec1812f95	8fea87303bc95c6f
54	76b4448cdf78e95b	dea048283fd97502	fecd331edbf976c5	276abdeaeae9850f	71c48267ea7f7c34	71b8ace06ac6c570	30b2ad1d7228bff8	f741f38ec1812f95
55	1a73b4b3b8ebae51	76b4448cdf78e95b	dea048283fd97502	fecd331edbf976c5	694bd45ec3d78cb6	71c48267ea7f7c34	71b8ace06ac6c570	30b2ad1d7228bff8
56	985eff85858a8f60	1a73b4b3b8ebae51	76b4448cdf78e95b	dea048283fd97502	308a14225df6d754	694bd45ec3d78cb6	71c48267ea7f7c34	71b8ace06ac6c570
57	df9839c71003605e	985eff85858a8f60	1a73b4b3b8ebae51	76b4448cdf78e95b	71004c9a4f292bc9	308a14225df6d754	694bd45ec3d78cb6	71c48267ea7f7c34
58	6ba55028943349c4	df9839c71003605e	985eff85858a8f60	1a73b4b3b8ebae51	fde7fbf2ef45e25c	71004c9a4f292bc9	308a14225df6d754	694bd45ec3d78cb6

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  58 | 6ba55028943349c4  df9839c71003605e  985eff85858a8f60  1a73b4b3b8ebae51  fde7fbf2ef45e25c  71004c9a4f292bc9  308a14225df6d754  694bd45ec3d78cb6
  59 | 3ea837cbb14666e7  6ba55028943349c4  df9839c71003605e  985eff85858a8f60  466f4a55adc3afd1  fde7fbf2ef45e25c  71004c9a4f292bc9  308a14225df6d754
  60 | 9d1be3c2af929a3c  3ea837cbb14666e7  6ba55028943349c4  df9839c71003605e  467d3674b9c5a7b0  466f4a55adc3afd1  fde7fbf2ef45e25c  71004c9a4f292bc9
  61 | 6fbf656f857ababb  9d1be3c2af929a3c  3ea837cbb14666e7  6ba55028943349c4  381c0af950a8dd18  467d3674b9c5a7b0  466f4a55adc3afd1  fde7fbf2ef45e25c
  62 | 651cd99b64e61c15  6fbf656f857ababb  9d1be3c2af929a3c  3ea837cbb14666e7  2706d18afcffb9b1  381c0af950a8dd18  467d3674b9c5a7b0  466f4a55adc3afd1
  63 | 84bea3cc875a5678  651cd99b64e61c15  6fbf656f857ababb  9d1be3c2af929a3c  b4b56f61706d8cbf  2706d18afcffb9b1  381c0af950a8dd18  467d3674b9c5a7b0
  64 | 4d34d74ff8a0c6dd  84bea3cc875a5678  651cd99b64e61c15  6fbf656f857ababb  b714928886ec5964  b4b56f61706d8cbf  2706d18afcffb9b1  381c0af950a8dd18
  65 | 32fa6c55c2c26ff8  4d34d74ff8a0c6dd  84bea3cc875a5678  651cd99b64e61c15  71a7e00728c0abd4  b714928886ec5964  b4b56f61706d8cbf  2706d18afcffb9b1
  66 | 2b2e6758cf53f8c5  32fa6c55c2c26ff8  4d34d74ff8a0c6dd  84bea3cc875a5678  947879039dfa2351  71a7e00728c0abd4  b714928886ec5964  b4b56f61706d8cbf
  67 | 4e6b51103b47da01  2b2e6758cf53f8c5  32fa6c55c2c26ff8  4d34d74ff8a0c6dd  3075b7ec85639108  947879039dfa2351  71a7e00728c0abd4  b714928886ec5964
  68 | c56203707ca0609d  4e6b51103b47da01  2b2e6758cf53f8c5  32fa6c55c2c26ff8  4d519c4ef24a18d4  3075b7ec85639108  947879039dfa2351  71a7e00728c0abd4
  69 | fbfc3c234caf2ddc  c56203707ca0609d  4e6b51103b47da01  2b2e6758cf53f8c5  a9b513ee733b3172  4d519c4ef24a18d4  3075b7ec85639108  947879039dfa2351
  70 | a0112f81127c109f  fbfc3c234caf2ddc  c56203707ca0609d  4e6b51103b47da01  9c953cc62c5d6b83  a9b513ee733b3172  4d519c4ef24a18d4  3075b7ec85639108
  71 | bcdce8d21327ab54  a0112f81127c109f  fbfc3c234caf2ddc  c56203707ca0609d  c85d14c41b6cd30a  9c953cc62c5d6b83  a9b513ee733b3172  4d519c4ef24a18d4
  72 | 1b988e67cd7c9358  bcdce8d21327ab54  a0112f81127c109f  fbfc3c234caf2ddc  1589548cec6dbb55  c85d14c41b6cd30a  9c953cc62c5d6b83  a9b513ee733b3172
  73 | 75c30d0387c1ee3f  1b988e67cd7c9358  bcdce8d21327ab54  a0112f81127c109f  ce19b1f2b19a93c6  1589548cec6dbb55  c85d14c41b6cd30a  9c953cc62c5d6b83
  74 | 2f7115748a70f60b  75c30d0387c1ee3f  1b988e67cd7c9358  bcdce8d21327ab54  f6c89d0d15dbc5ec  ce19b1f2b19a93c6  1589548cec6dbb55  c85d14c41b6cd30a
  75 | 95183b1ae81e7222  2f7115748a70f60b  75c30d0387c1ee3f  1b988e67cd7c9358  a59c59a94141371c  f6c89d0d15dbc5ec  ce19b1f2b19a93c6  1589548cec6dbb55
  76 | 04ac4102579d83a2  95183b1ae81e7222  2f7115748a70f60b  75c30d0387c1ee3f  dcc33011e8d991b1  a59c59a94141371c  f6c89d0d15dbc5ec  ce19b1f2b19a93c6
  77 | 35b01d13e2329026  04ac4102579d83a2  95183b1ae81e7222  2f7115748a70f60b  bdaab7350ff1335c  dcc33011e8d991b1  a59c59a94141371c  f6c89d0d15dbc5ec
  78 | c5e160f1904ab888  35b01d13e2329026  04ac4102579d83a2  95183b1ae81e7222  e3389a089e966631  bdaab7350ff1335c  dcc33011e8d991b1  a59c59a94141371c
  79 | 5ad2871d518323b9  c5e160f1904ab888  35b01d13e2329026  04ac4102579d83a2  66b0cd44e8a16f55  e3389a089e966631  bdaab7350ff1335c  dcc33011e8d991b1

Final Hash Value: 847d4ec5ef2ddb00738e8bf3483b572cc68add191e3857594d6b7216fb802f62eaa76a7ba6683a2800cb0b2aa3f1f5f6bf3e3a1858f4dbd63c2f39e443ba4101
```

Input Test Case 3: >896 bits(>112 characters)

```
PS E:\aa vit sem6\z_Crypto-lab\Ex8_sha512> cd "e:\aa vit sem6\z_Crypto-lab\Ex8_sha512\" ; if ($?) { javac sha512.java } ; if ($?) { java sha512 }
Enter Input: arkitabaruaarkitabaruaarkitabaruaarkitabaruaarkitabaruaarkitabaruaarkitabaruaarkitabaruacryptographycryptographyarkitaba
rua
No of characters in input: 123

========================================== Processing Block 1 ==========================================
Round |        A                B                C                D                E                F                G
        H
-----------------------------------------------------------------------------------------------------------------------------------
  00  | f6bfd6a2315e4056  6a09e667f3bcc908  bb67ae8584caa73b  3c6ef372fe94f82b  58db0a1def1681f2  510e527fade682d1  9b05688c2b3e6c1f  1f83d9abfb41bd6b
  01  | 578270f88046c4e3  f6bfd6a2315e4056  6a09e667f3bcc908  bb67ae8584caa73b  bbf053c9513d1bce  58db0a1def1681f2  510e527fade682d1  9b05688c2b3e6c1f
  02  | 247211265415cabe  578270f88046c4e3  f6bfd6a2315e4056  6a09e667f3bcc908  93d05644bc191a60  bbf053c9513d1bce  58db0a1def1681f2  510e527fade682d1
  03  | 9ef1a7932487d721  247211265415cabe  578270f88046c4e3  f6bfd6a2315e4056  f9574a8055786d25  93d05644bc191a60  bbf053c9513d1bce  58db0a1def1681f2
  04  | 357f739e99f258cc  9ef1a7932487d721  247211265415cabe  578270f88046c4e3  6e37900efbacdee6  f9574a8055786d25  93d05644bc191a60  bbf053c9513d1bce
  05  | f56748716c57e599  357f739e99f258cc  9ef1a7932487d721  247211265415cabe  140b87921544b3e7  6e37900efbacdee6  f9574a8055786d25  93d05644bc191a60
  06  | e462ad8780aaf0e5  f56748716c57e599  357f739e99f258cc  9ef1a7932487d721  d6aa4be88f214a80  140b87921544b3e7  6e37900efbacdee6  f9574a8055786d25
  07  | bacff71681f5d80c  e462ad8780aaf0e5  f56748716c57e599  357f739e99f258cc  7ed54d8b2890081e  d6aa4be88f214a80  140b87921544b3e7  6e37900efbacdee6
  08  | 80a09f26ae0017fc  bacff71681f5d80c  e462ad8780aaf0e5  f56748716c57e599  2f9508352bc16cf8  7ed54d8b2890081e  d6aa4be88f214a80  140b87921544b3e7
  09  | 21bb2ef8db463522  80a09f26ae0017fc  bacff71681f5d80c  e462ad8780aaf0e5  6f623e4af3dcb532  2f9508352bc16cf8  7ed54d8b2890081e  d6aa4be88f214a80
  10  | 949256ccbdaac8dd  21bb2ef8db463522  80a09f26ae0017fc  bacff71681f5d80c  6544f21cdb725228  6f623e4af3dcb532  2f9508352bc16cf8  7ed54d8b2890081e
  11  | 66db3dc19dc24e9f  949256ccbdaac8dd  21bb2ef8db463522  80a09f26ae0017fc  343bab6bc98d32da  6544f21cdb725228  6f623e4af3dcb532  2f9508352bc16cf8
  12  | 024a5d3acffac232  66db3dc19dc24e9f  949256ccbdaac8dd  21bb2ef8db463522  25dfbecc556fe87d  343bab6bc98d32da  6544f21cdb725228  6f623e4af3dcb532
  13  | 47357002e3b030e6  024a5d3acffac232  66db3dc19dc24e9f  949256ccbdaac8dd  2848db0a60d7cb11  25dfbecc556fe87d  343bab6bc98d32da  6544f21cdb725228
  14  | bf437eab0f707824  47357002e3b030e6  024a5d3acffac232  66db3dc19dc24e9f  8751f638cf14c77d  2848db0a60d7cb11  25dfbecc556fe87d  343bab6bc98d32da
  15  | 2373c8fa67af0ff9  bf437eab0f707824  47357002e3b030e6  024a5d3acffac232  20460ddea2e73a8b  8751f638cf14c77d  2848db0a60d7cb11  25dfbecc556fe87d
  16  | 55c5e013d9875f53  2373c8fa67af0ff9  bf437eab0f707824  47357002e3b030e6  18c86230b42b4e98  20460ddea2e73a8b  8751f638cf14c77d  2848db0a60d7cb11
  17  | c23f6d200c0a5ad3  55c5e013d9875f53  2373c8fa67af0ff9  bf437eab0f707824  8889c808708e394f  18c86230b42b4e98  20460ddea2e73a8b  8751f638cf14c77d
  18  | 87fea4f313539cbb  c23f6d200c0a5ad3  55c5e013d9875f53  2373c8fa67af0ff9  800b0c2e60373d7a  8889c808708e394f  18c86230b42b4e98  20460ddea2e73a8b
  19  | 275fecc549cb1bf4  87fea4f313539cbb  c23f6d200c0a5ad3  55c5e013d9875f53  6b0845dcaf5cb1d6  800b0c2e60373d7a  8889c808708e394f  18c86230b42b4e98
  20  | 2607abf90cffe4de  275fecc549cb1bf4  87fea4f313539cbb  c23f6d200c0a5ad3  affcafa348d4a082  6b0845dcaf5cb1d6  800b0c2e60373d7a  8889c808708e394f
  21  | 7fb82e77bf7e36f5  2607abf90cffe4de  275fecc549cb1bf4  87fea4f313539cbb  9bbfa38a6bf59a91  affcafa348d4a082  6b0845dcaf5cb1d6  800b0c2e60373d7a
```

22	5deb1c165188914a	7fb82e77bf7e36f5	2607abf90cffe4de	275fecc549cb1bf4	c787f38d08495a57	9bbfa38a6bf59a91	affcafa348d4a082	6b0845dcaf5cb1d6
23	a15da650eb3c06d9	5deb1c165188914a	7fb82e77bf7e36f5	2607abf90cffe4de	a8ca42c939979d97	c787f38d08495a57	9bbfa38a6bf59a91	affcafa348d4a082
24	ee443023feb35b3b	a15da650eb3c06d9	5deb1c165188914a	7fb82e77bf7e36f5	6d7999a441a7716a	a8ca42c939979d97	c787f38d08495a57	9bbfa38a6bf59a91
25	1b67817ea8a28b00	ee443023feb35b3b	a15da650eb3c06d9	5deb1c165188914a	5a6e781d045a2843	6d7999a441a7716a	a8ca42c939979d97	c787f38d08495a57
26	a3fd159f832b0b79	1b67817ea8a28b00	ee443023feb35b3b	a15da650eb3c06d9	7951396b396a58f3	5a6e781d045a2843	6d7999a441a7716a	a8ca42c939979d97
27	5968c7249072d805	a3fd159f832b0b79	1b67817ea8a28b00	ee443023feb35b3b	61e538e32ca1ecf1	7951396b396a58f3	5a6e781d045a2843	6d7999a441a7716a
28	c0a86e88669da3d8	5968c7249072d805	a3fd159f832b0b79	1b67817ea8a28b00	296d45891ac9e204	61e538e32ca1ecf1	7951396b396a58f3	5a6e781d045a2843
29	5a21947517579d17	c0a86e88669da3d8	5968c7249072d805	a3fd159f832b0b79	52f06029b290f8c5	296d45891ac9e204	61e538e32ca1ecf1	7951396b396a58f3
30	99283d4aaf8ee293	5a21947517579d17	c0a86e88669da3d8	5968c7249072d805	0a79f3bbc23cf393	52f06029b290f8c5	296d45891ac9e204	61e538e32ca1ecf1
31	0010dbc41273eec2	99283d4aaf8ee293	5a21947517579d17	c0a86e88669da3d8	bafeda48274b97b2	0a79f3bbc23cf393	52f06029b290f8c5	296d45891ac9e204
32	5fa23ec2bc86c3fc	0010dbc41273eec2	99283d4aaf8ee293	5a21947517579d17	5ca31fba06c2ce3a	bafeda48274b97b2	0a79f3bbc23cf393	52f06029b290f8c5
33	79aeb9d5136d7cd3	5fa23ec2bc86c3fc	0010dbc41273eec2	99283d4aaf8ee293	d87b8bca56ca0f32	5ca31fba06c2ce3a	bafeda48274b97b2	0a79f3bbc23cf393
34	8c49936ac87ba49e	79aeb9d5136d7cd3	5fa23ec2bc86c3fc	0010dbc41273eec2	f3a4cbf6832f2a08	d87b8bca56ca0f32	5ca31fba06c2ce3a	bafeda48274b97b2
35	1169d1c5c5e8c34d	8c49936ac87ba49e	79aeb9d5136d7cd3	5fa23ec2bc86c3fc	d39b9a40e5590be1	f3a4cbf6832f2a08	d87b8bca56ca0f32	5ca31fba06c2ce3a
36	876c920d264fade3	1169d1c5c5e8c34d	8c49936ac87ba49e	79aeb9d5136d7cd3	292769855886d172	d39b9a40e5590be1	f3a4cbf6832f2a08	d87b8bca56ca0f32
37	726dd83c4ed6475e	876c920d264fade3	1169d1c5c5e8c34d	8c49936ac87ba49e	2f8d54a84dbb40ed	292769855886d172	d39b9a40e5590be1	f3a4cbf6832f2a08
38	14418fbc0a419560	726dd83c4ed6475e	876c920d264fade3	1169d1c5c5e8c34d	06d10a5b854c4434	2f8d54a84dbb40ed	292769855886d172	d39b9a40e5590be1
39	6c195ef654c99de4	14418fbc0a419560	726dd83c4ed6475e	876c920d264fade3	8877e60c8b4ac09f	06d10a5b854c4434	2f8d54a84dbb40ed	292769855886d172
40	554678d2e6ee4f48	6c195ef654c99de4	14418fbc0a419560	726dd83c4ed6475e	3367021fac30dd62	8877e60c8b4ac09f	06d10a5b854c4434	2f8d54a84dbb40ed
41	1154e253e8cc6ce7	554678d2e6ee4f48	6c195ef654c99de4	14418fbc0a419560	bcee9fd01f3c76fa	3367021fac30dd62	8877e60c8b4ac09f	06d10a5b854c4434
42	ee4e2e8b5462c3d7	1154e253e8cc6ce7	554678d2e6ee4f48	6c195ef654c99de4	dc16f5e35a9e56e5	bcee9fd01f3c76fa	3367021fac30dd62	8877e60c8b4ac09f
43	c83c361835d5146d	ee4e2e8b5462c3d7	1154e253e8cc6ce7	554678d2e6ee4f48	5972e22df5026340	dc16f5e35a9e56e5	bcee9fd01f3c76fa	3367021fac30dd62
44	80e4424b1fdcb464	c83c361835d5146d	ee4e2e8b5462c3d7	1154e253e8cc6ce7	ad7eeb1367aaaa5c	5972e22df5026340	dc16f5e35a9e56e5	bcee9fd01f3c76fa
45	073a702ccb3fa012	80e4424b1fdcb464	c83c361835d5146d	ee4e2e8b5462c3d7	a41f5a3bb6ba7bed	ad7eeb1367aaaa5c	5972e22df5026340	dc16f5e35a9e56e5
46	757cf2630bafabf0	073a702ccb3fa012	80e4424b1fdcb464	c83c361835d5146d	0aeb188169ccd13c	a41f5a3bb6ba7bed	ad7eeb1367aaaa5c	5972e22df5026340
47	de00d2960bc2292a	757cf2630bafabf0	073a702ccb3fa012	80e4424b1fdcb464	e2ba8b968b5cb9db	0aeb188169ccd13c	a41f5a3bb6ba7bed	ad7eeb1367aaaa5c
48	209905de2e5a897d	de00d2960bc2292a	757cf2630bafabf0	073a702ccb3fa012	377eb94e4056784f	e2ba8b968b5cb9db	0aeb188169ccd13c	a41f5a3bb6ba7bed
49	4026d7597d830ab4	209905de2e5a897d	de00d2960bc2292a	757cf2630bafabf0	00e5f4108100d2b0	377eb94e4056784f	e2ba8b968b5cb9db	0aeb188169ccd13c
50	1f263f8dcf999adb	4026d7597d830ab4	209905de2e5a897d	de00d2960bc2292a	1ef6ea1630a2afa0	00e5f4108100d2b0	377eb94e4056784f	e2ba8b968b5cb9db
51	7091e7bb70d3e91b	1f263f8dcf999adb	4026d7597d830ab4	209905de2e5a897d	bc8baa4189664c08	1ef6ea1630a2afa0	00e5f4108100d2b0	377eb94e4056784f
52	f91487d9589858cc	7091e7bb70d3e91b	1f263f8dcf999adb	4026d7597d830ab4	219ac9fa25843618	bc8baa4189664c08	1ef6ea1630a2afa0	00e5f4108100d2b0
53	5405f716dc5fa8d2	f91487d9589858cc	7091e7bb70d3e91b	1f263f8dcf999adb	ae057c89ca49653f	219ac9fa25843618	bc8baa4189664c08	1ef6ea1630a2afa0
54	2e3d073dec0e73b3	5405f716dc5fa8d2	f91487d9589858cc	7091e7bb70d3e91b	7df786eff0d61e6e	ae057c89ca49653f	219ac9fa25843618	bc8baa4189664c08
55	bc346921b28312ad	2e3d073dec0e73b3	5405f716dc5fa8d2	f91487d9589858cc	f0740e85f9353ad7	7df786eff0d61e6e	ae057c89ca49653f	219ac9fa25843618
56	d1c524d9230b3464	bc346921b28312ad	2e3d073dec0e73b3	5405f716dc5fa8d2	86af5d26886079fc	f0740e85f9353ad7	7df786eff0d61e6e	ae057c89ca49653f
57	7b337595714e0bdc	d1c524d9230b3464	bc346921b28312ad	2e3d073dec0e73b3	48ccaa35ca21f31c	86af5d26886079fc	f0740e85f9353ad7	7df786eff0d61e6e
58	544bd57e65c2d15e	7b337595714e0bdc	d1c524d9230b3464	bc346921b28312ad	2701d40409b84f4c	48ccaa35ca21f31c	86af5d26886079fc	f0740e85f9353ad7
59	a022565a2919bf30	544bd57e65c2d15e	7b337595714e0bdc	d1c524d9230b3464	d17d258e0d4b8b6e	2701d40409b84f4c	48ccaa35ca21f31c	86af5d26886079fc
60	1d0275ca16bd945c	a022565a2919bf30	544bd57e65c2d15e	7b337595714e0bdc	cf14958c8d1078cc	d17d258e0d4b8b6e	2701d40409b84f4c	48ccaa35ca21f31c
61	5b32d8bbed605577	1d0275ca16bd945c	a022565a2919bf30	544bd57e65c2d15e	48089cf8c9b706bf	cf14958c8d1078cc	d17d258e0d4b8b6e	2701d40409b84f4c
62	0d446f0fd6de92c6	5b32d8bbed605577	1d0275ca16bd945c	a022565a2919bf30	ede66d318410768f	48089cf8c9b706bf	cf14958c8d1078cc	d17d258e0d4b8b6e
63	5713ac541878cd08	0d446f0fd6de92c6	5b32d8bbed605577	1d0275ca16bd945c	52404f2d8f889402	ede66d318410768f	48089cf8c9b706bf	cf14958c8d1078cc
64	267b6fd3687e7774	5713ac541878cd08	0d446f0fd6de92c6	5b32d8bbed605577	bac8e7202e732d7e	52404f2d8f889402	ede66d318410768f	48089cf8c9b706bf
65	956fa246e9542abc	267b6fd3687e7774	5713ac541878cd08	0d446f0fd6de92c6	ee26f539f7d0d8d2	bac8e7202e732d7e	52404f2d8f889402	ede66d318410768f
66	1d837a95b34f3419	956fa246e9542abc	267b6fd3687e7774	5713ac541878cd08	7087321c172643e8	ee26f539f7d0d8d2	bac8e7202e732d7e	52404f2d8f889402
67	c4f67aa86f81dd5a	1d837a95b34f3419	956fa246e9542abc	267b6fd3687e7774	9358a924f130031b	7087321c172643e8	ee26f539f7d0d8d2	bac8e7202e732d7e
68	2a667888baf6b154	c4f67aa86f81dd5a	1d837a95b34f3419	956fa246e9542abc	07d7cc96ec5c13d7	9358a924f130031b	7087321c172643e8	ee26f539f7d0d8d2
69	ed8339251f8bf6f1	2a667888baf6b154	c4f67aa86f81dd5a	1d837a95b34f3419	e5690c6e486d91fd	07d7cc96ec5c13d7	9358a924f130031b	7087321c172643e8
70	1e1d458c936648cf	ed8339251f8bf6f1	2a667888baf6b154	c4f67aa86f81dd5a	5957c22ac6a92d2e	e5690c6e486d91fd	07d7cc96ec5c13d7	9358a924f130031b
71	d21355f952d2bc73	1e1d458c936648cf	ed8339251f8bf6f1	2a667888baf6b154	5d6684c16dfe75d7	5957c22ac6a92d2e	e5690c6e486d91fd	07d7cc96ec5c13d7
72	d1356e503e0a1353	d21355f952d2bc73	1e1d458c936648cf	ed8339251f8bf6f1	924dc3d2b3281b74	5d6684c16dfe75d7	5957c22ac6a92d2e	e5690c6e486d91fd
73	d2315934527aa6a3	d1356e503e0a1353	d21355f952d2bc73	1e1d458c936648cf	9e3fa6b21f0ab0f6	924dc3d2b3281b74	5d6684c16dfe75d7	5957c22ac6a92d2e
74	ead79ec8d9134a99	d2315934527aa6a3	d1356e503e0a1353	d21355f952d2bc73	db33500c89e1353b	9e3fa6b21f0ab0f6	924dc3d2b3281b74	5d6684c16dfe75d7
75	3a688f137f42a7ad	ead79ec8d9134a99	d2315934527aa6a3	d1356e503e0a1353	0384290e93e1bd8b	db33500c89e1353b	9e3fa6b21f0ab0f6	924dc3d2b3281b74
76	1ed42508f240dae6	3a688f137f42a7ad	ead79ec8d9134a99	d2315934527aa6a3	e8939dd1229243a3	0384290e93e1bd8b	db33500c89e1353b	9e3fa6b21f0ab0f6
77	75bd762221d0e61b	1ed42508f240dae6	3a688f137f42a7ad	ead79ec8d9134a99	03a126dfd23dd08a	e8939dd1229243a3	0384290e93e1bd8b	db33500c89e1353b
78	f78afdff932b1e60	75bd762221d0e61b	1ed42508f240dae6	3a688f137f42a7ad	d22c7bc848aa3d04	03a126dfd23dd08a	e8939dd1229243a3	0384290e93e1bd8b
79	b56c26854e129291	f78afdff932b1e60	75bd762221d0e61b	1ed42508f240dae6	4edac8c29571e65f	d22c7bc848aa3d04	03a126dfd23dd08a	e8939dd1229243a3

```
========================================== Processing Block 2 ==========================================
Round |        A                 B                 C                 D                 E                 F                 G
        H
-----------------------------------------------------------------------------------------------------------------------------------------------------------------------
  00  | 1c4fadfd14a8abee  1f760ced41cf5b99  b2f2ac8517f5c59b  b22c69952065de46  97f7a9fae19d3c34  9fe91b4243586930  6d31e45473e8a923  2325008bcd7f8df5
  01  | eaedb95b5285da4c  1c4fadfd14a8abee  1f760ced41cf5b99  b2f2ac8517f5c59b  8919b0b5bd2c4102  97f7a9fae19d3c34  9fe91b4243586930  6d31e45473e8a923
  02  | 724223fdd1942001  eaedb95b5285da4c  1c4fadfd14a8abee  1f760ced41cf5b99  bc6b489c8b46e93f  8919b0b5bd2c4102  97f7a9fae19d3c34  9fe91b4243586930
  03  | 770520f6e8e5b20d  724223fdd1942001  eaedb95b5285da4c  1c4fadfd14a8abee  85a7638f5fda2ff5  bc6b489c8b46e93f  8919b0b5bd2c4102  97f7a9fae19d3c34
  04  | 9d078127346bd4f1  770520f6e8e5b20d  724223fdd1942001  eaedb95b5285da4c  6d5e85f4411181c0  85a7638f5fda2ff5  bc6b489c8b46e93f  8919b0b5bd2c4102
  05  | 5259c2a5da0ce393  9d078127346bd4f1  770520f6e8e5b20d  724223fdd1942001  0070ec7ce7a910a6  6d5e85f4411181c0  85a7638f5fda2ff5  bc6b489c8b46e93f
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Final Hash Value: 8310f1cc5703f948c443f5bc8a1835cbea1d416c81a7d32ecfbf7ba058646552c0dfcf3d9f730631cf8ef11053c12e1ba72fe45ec18491799bad895a9438eeee
PS E:\aa vit sem6\z_Crypto-lab\Ex8_sha512> ^B
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